Hydrology

Vrije Universiteit Amsterdam - Faculty of Science - Hydrology - 2025-2026

The Hydrology master's programme provides students with sound scientific knowledge of how water cycles through the Earth’s atmosphere, surface and groundwater systems and how water quantity and quality are modified due to natural processes or in response to human interference with the water cycle (e.g. pollution, land use change, etc.). Key questions addressed include: How does the hydrological system function and how does it relate with climate, vegetation and society? How can we ensure there is enough water available for every person? Can we ensure water quality for agriculture, biodiversity and drinking water? How can we minimise the impact of extreme flood events in densely populated urban areas?

In the first year, the Hydrology MSc programme consists of a combination of lectures, computer practicals, and field courses in The Netherlands and in Luxembourg. This year is used to teach you the basics of theoretical and experimental hydrology, methods to assess hydrological risk for society, and apply knowledge of hydrological processes to real world water management issues. In the second year, you will delve into state-the-art methods. Moreover, this year allows further specialisation in your area of interest, and is concluded by a 6 months Master's thesis project.

Info

Level	Master
Language	English
Duration	2 years
Type	Full Time
Credits	120 EC
Faculty	Faculty of Science

Teaching and examination regulation 2025-2026
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Hydrology year 1

Description

MSc Hydrology is a two-year program. The first year of the program consists completely of compulsory courses that are distributed over the two teaching lines: (1) Process Hydrology and (2) Water Risk. The students get acquainted with several key components of the hydrological system (e.g. in the coruses Catchment Hydrology, Ecohydrology, and Groundwater Processes), as well as their links to society (e.g. in Water Economics, and Integrated Modelling in Hydrology). At the end of the first year, the students learn how to perform hydrological measurements (1-week field trip as part of the Research Techniques in Hydrology course), and then apply these themselves in a field study (including 3-4 week field trip as part of the Field Research Project Hydrology course) which is aimed at teaching the students to collect data themselves, formulate and find answers to hypothesis, and link the various subsystems in hydrology and their ties to societal issues.

Courses

Course Name	Period	Credits	Code
Catchment Hydrology	P1	6EC	AM_1278
Ecohydrology	P1	6EC	AM_450014
Catchment Biogeochemistry	P2	6EC	AM_1279
Climate Hydrological Processes	P2	6EC	AM_1196
Integrated Modeling in Hydrology	P3	6EC	AM_1165
Groundwater Processes	P4	6EC	AM_1164
Water Economics	P4	6EC	AM_1167
Measuring Techniques in Hydrology	P5	6EC	AM_1315
Field Research Project Hydrology	P5+6	12EC	AM_1314

Hydrology year 2

Description

The 2nd year starts with two compulsory courses (Frontiers in Hydrology and Water Risks). These are advances courses that build upon some courses from the first year.

After period 1, there is room for two fulltime elective courses (12 ECTS). This allows you to specialize in your area of interest. The elective options can be filled with MSc courses shared with other VU programs (such as ES, GECP and ERM); this includes courses such as Water Politics and Governance, Environmental Policy for Global Environmental Change and Policy, Climate Modelling, Environmental Remote Sensing, Advanced Spatial Analysis, Global Biogeochemical Cycles, and Plastic and Chemical Pollution. Alternatively, you can elect courses offered by partner universities (such as UvA, IHE Delft, TU Delft).

The other compulsory course is the Master Thesis Hydrology (36 ECTS). The planning of this course is somewhat flexible; the student works on their thesis research, and follows elective courses whenever they are given. This means that the student can start with working (parttime) on his/her thesis research in period 2.

Programme Sections

- Hydrology year 2 elective courses
- Hydrology year 2 compulsory courses

Hydrology year 2 elective courses

Description

In the 2nd year of the MSc Hydrology, 12 ECTS should be spent on elective courses. These can include advanced courses in Hydrology, courses from related MSc programs (such as GECP and ERM), or even courses offered at partner universities (such as UvA). Note that in case you are interested in following a course that is not listed here, you need to contact the coordinator after whose approval a request can be sent to the examination committee for the student to include that course in their programme.

Courses

Course Name	Period	Credits	Code

Course Name	Period	Credits	Code
Advanced Spatial Analyses	P2	6EC	AM_1197
Biological Oceanography	P2	6EC	AMU_0021
Environmental Policy for GEC&P	P2	6EC	AM_1241
Plastic and Chemical Pollution	P2	6EC	AM_1259
Climate Modelling	P3	6EC	AM_450004
Environmental Remote Sensing	P3	6EC	AM_450145
Water Politics and Governance	P3	6EC	AM_1192
Global Biogeochemical Cycles	P4	6EC	AM_450332
Reflection Seismic for Geologists	P4	6EC	AM_450170

Hydrology year 2 compulsory courses

Description

The 2nd year of the Hydrology MSc programme has three compulsory courses: Frontiers in Hydrology, Water Risks, and the Master Thesis Hydrology (36 ECTS). The remaining 12 ECTS are filled with elective courses.

Courses

Course Name	Period	Credits	Code
Master Thesis Hydrology	Ac. Year (September)	36EC	AM_1170
Frontiers in Hydrology	P1	6EC	AM_1280
Water Risks	P1	6EC	AM_1210

Courses

Advanced Spatial Analyses

Course Code	AM_1197
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.A. de Bruijn
Examiner	dr. J.A. de Bruijn
Teaching Staff	dr. J.A. de Bruijn, dr. N.A.J. Schutgens
Teaching method(s)	Study Group, Computer lab, Written partial exam, Lecture

Course Objective

The objective of this course is to become familiar with more advanced approaches for the analysis of (large) spatial and temporal datasets in the context of earth and environmental sciences, for example related to (addressing) climate change.

Course Content

In this course you will acquire the necessary skills to conduct spatial-temporal analysis in the context of earth and environmental sciences. The course starts with improving more fundamental programming in Python, and gradually focuses on tools that are useful for spatial-temporal analyses: reading and writing spatial and time-series data, libraries for handling spatial data and time series data, data analysis, and visualization of results. You will learn these skills using several practical cases, such as making climate stripes, simulating wildfires, optimization of an energy-grid using weather data, and using machine learning to detect flood events in satellite imagery. The gained knowledge in each exercise should be transferable across the domain of earth sciences.

Additional Information Teaching Methods

Each week will consist of one or two lectures and multiple computer labs (tutorials) where students can work on the

assignments. Teachers and/or teaching assistants will be available to assist.

Method of Assessment

The course requires 8 weeks of part-time (50%) study, and students are expected to spend approximately 20 hours a week on this course.

Entry Requirements

This course assumes a basic knowledge of GIS and GIS-based spatial analysis (such as the course GIS and Digital Spatial Data (AB_1076), or GIS en Aardobservatie AB_1271 or equivalent), as well as basic knowledge of Python. This requirement is met by all students who completed their BSc in Earth Sciences, or Earth, Economics and Sustainability at VU University. MSc students in Econometrics and Operations are not expected to experience issues when they do not have this background, especially when they are experienced in other programming languages.

Additional guidance will be provided to brush up Python knowledge for students who need it.

Literature

Selected literature will be provided on Canvas.

Additional Information Target Audience

MSc students in Earth Sciences, MSc students in Hydrology, MSc students in Econometrics and Operations Research with specialization Climate Econometrics, and other relevant studies.

Additional Information

It is advisable for students to have their own laptop for computer exercises.

Explanation Canvas

We will use canvas as well as a custom website to provide information and assignments throughout the course. Assignments will be submitted through canvas.

Biological Oceanography

Course Code	AMU_0021
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.E. Vonk
Examiner	dr. J.E. Vonk
Teaching Staff	
Teaching method(s)	

Course Objective

- Identify main biological oceanographic processes (4) and describe their interdependence
- Specify and differentiate between biological processes involved in the biological pump
- Survey relevant literature about a given research topic within biological oceanography and summarize in an abstract
- Develop and formulate a follow-up research question / objective on a given topic within biological oceanography and describe how the research should be conducted
- Present a lecture aimed at explaining a research topic to a scientific audience and proposing a continuation research project
- Explain the importance of the biology of the ocean for global change developments
- Critically discuss, oppose and defend scientific issues within the field of biological oceanography

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130120

Additional Information Teaching Methods

- Lecture
- Seminar
- Computer lab session/practical training
- Presentation/symposium
- Self-study
- Supervision/feedback meeting
- Fieldwork/excursion
- Working independently on e.g. a project or thesis

Method of Assessment

- Exam
- Abstract and formulation research objective
- Oral presentation

Literature

(**not** obligatory) the book: Oceanography and Marine Biology. An Introduction to Marine Science. (D.W. Townsend. Sinauer Associates, ISBN 9780878936021) may serve as introduction to the course.

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Catchment Biogeochemistry

Course Code	AM_1279
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.E. Vonk
Examiner	dr. J.E. Vonk
Teaching Staff	dr. J.E. Vonk, dr. F.M. Béen
Teaching method(s)	Lecture

Course Objective

The main objective of Catchment Biogeochemistry is to provide students with a basic understanding of the landscape controls on water biogeochemistry. Streams and rivers serve as integrators of their surroundings and are therefore a diagnostic tool to assess the properties and ongoing (natural or anthropogenic) changes of the land that they drain.
More specifically, the student will at the end of the course:

- have acquired a qualitative and quantitative understanding of biogeochemical processes in the soil that influence the biogeochemical composition of water;
- understand how the biogeochemical composition of water changes within the hydrological cycle of a catchment: from precipitation, to groundwater to surface water;
- understand, describe and utilize techniques for assessing biogeochemical constituents in water, and be able to interpret these data;
- be familiar with different types of pollutants affecting water quality, and understand how pollutants enter, move through, and interact with the hydrological system.
- be able to independently interpret, visualize, and discuss existing catchment biogeochemistry data by means of development and presentation of a conference poster
- be able to comprehend and present scientific articles on catchment biogeochemistry, and be able to criticise and discuss its content

Course Content

The biogeochemistry and quality of water on Earth is a function of processes that occur in the human and natural environment. Water of sufficient quality is required for, for example, agricultural and domestic usage, as well as for healthy ecosystems. Knowledge on water biogeochemistry is needed to study and interpret the quality of ground and surface water.

The following course topics are included: water sampling, analysis and assessment of water and its biogeochemical constituents, isotopic composition of water and its constituents, processes that act upon biogeochemical constituents in water (e.g. weathering, degradation), carbonate chemistry, dynamics and transport of pollutants.

Additional Information Teaching Methods

The course consists of lectures, supplemented with practicals (assignment), group discussions, and a lab afternoon.

Method of Assessment

Written examination (40%),
assignment (40%),
discussions (20%).
One has to pass all components (higher than 5.5), no compensation possible.

Additional Information Target Audience

Hydrology MSc students and other Earth Science-related MSc programs

Catchment Hydrology

Course Code	AM_1278
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. W.R. Berghuijs MSc
Examiner	dr. W.R. Berghuijs MSc
Teaching Staff	dr. W.R. Berghuijs MSc
Teaching method(s)	Seminar, Computer lab, Lecture

Course Objective

The objective of this course is to provide students with scientific theory, tools and methods for understanding and evaluating the surface water response of catchments to incoming precipitation. This requires knowledge about processes regulating the flow of water on the land surface and in river channels, the techniques for quantification of surface water flows and statistical methods for predicting extreme runoff events. In addition, experience with surface water flow modelling for predicting the behavior of rivers under different land use or climate conditions will be acquired.

The course contributes to the Knowledge and Understanding and Application of Knowledge and Understanding final attainment levels of the MSc Hydrology Program. Knowledge and understanding is obtained through studying theory as provided in the textbook, during the oral lectures and through self-study of scientific papers on rainfall-runoff response topics. Knowledge and understanding is applied to a rainfall-runoff model which simulations will be evaluated with measured data.

Course Content

The course consists of three main topics. We start with runoff processes leading to stormflow. This is followed by lectures on different hydrological modelling practices, and goodness-of-fit criteria. Finally, an overview of hydrodynamic and hydraulic theory that governs flow in open channels is given as well as an overview of discharge measuring techniques.

Topics are hillslope hydrology, hydrograph analysis, statistical methods to describe and quantify spatial and temporal variation in catchment runoff and reservoir and flow routing and scaling in hydrology. The spectrum of available models for runoff modelling, from classical lumped models to data-demanding distributed, physically-based hydrological models, will also be discussed. Finally, theory and understanding will be applied in a series of

modelling exercises applying a rainfall – runoff model to simulate runoff.

Additional Information Teaching Methods

The tuition consists of eleven classroom lectures and several computer modelling workshop sessions. The number of contact hours is in the order of 42.

Method of Assessment

The assessment is through a written exam (70%), an assessment of a selection of exercises (10%), and the HBV modeling assignment report (20%). Grades of all assessments should be at least a 5.5.

Entry Requirements

The student should be familiar with the subjects of the BSc course Introduction to Hydrology (AB_1074)

Literature

Main Textbook:

- S.L. Dingman, 2015. Physical hydrology, 3rd edition. ISBN: 978-1-4786-1118-9

Articles and other material (subject to updates)

- Seibert, J. & Vis, M. J. P. Teaching hydrological modeling with a user-friendly catchment-runoff-model software package. (Links to an external site.)
- Hydrology and Earth System Sciences, 2012, 16, 3315-3325 (Links to an external site.)
- Selection of Reader "Hydrology of catchments, rivers and deltas. Download Hydrology of catchments, rivers and deltas ", TUDelft, 2016
- Kirchner, J. W. Catchments as simple dynamical systems: Catchment characterization, rainfall-runoff modeling, and doing hydrology backward. (Links to an external site.)Water Resources Research, 2009, 45, W02429

Additional Information Target Audience

First-year M.Sc. Hydrology students, students from Earth Sciences, Earth and Economy or Natural Sciences M.Sc. programmes.

Additional Information

The course is open for participation to students from alternative M.Sc. programmes at the VU University Amsterdam, or from other universities.
If you are a professional and wish to attend this course you can also participate on a contract basis. In both cases please do contact the course coordinator to find out if you fulfill the background knowledge requirements and for enrolment procedures.

Custom Course Registration

The course is open for participation to students from alternative M.Sc. programs at the VU University Amsterdam, or from other universities. If you are a professional and wish to attend this course you can also participate on a contract basis. In both cases please do contact the course coordinator to find out if you fulfill the background knowledge requirements and for enrollment procedures.

Recommended background knowledge

The student should have good background knowledge of mathematics and physics at BSc level and have basic computer skills.
In addition, the student should have basic knowledge of Earth Science, as provided by the System Earth course (AB_450067).

Climate Hydrological Processes

Course Code	AM_1196
Credits	6
Period	P2
Course Level	400

Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. D. Coumou
Examiner	prof. dr. D. Coumou
Teaching Staff	
Teaching method(s)	Seminar, Computer lab

Course Objective

Present-day climate change is already increasing the intensity and/or frequency of extreme weather around the globe. A key challenge for environmental scientists today is to assess the likely societal impacts from future changes in hydro-meteorological extremes, with heavy rainfall leading to flooding and persistent dry conditions leading to drought-related impacts. To do such impact assessments, scientists have to rely on global climate models in combination with local hydro-meteorological impact models. This process is associated with substantial uncertainty for example because those models operate at different spatial resolution. To be able to properly interpret future impact studies, one needs to understand the most relevant physical processes of the climate system and how those processes are represented in climate models. This will be the focus of this course.

This course aims to provide students a solid understanding of the climate system including the most important underlying physical processes, its overall characteristics and its variability. The focus will be towards the intimate links between the climate system, the hydrological cycle and the large-scale atmospheric circulation. We will analyse which type of weather extremes have already increased in intensity and what kind of changes are to be expected for the near future.

Key goals for students to reach at the end of the course are:

1. Student can describe in own words how present-day climate change affects the hydrological cycle and how this impacts society.
2. Student can give a clear vision of the risks of climate change in his/her lifetime.
3. Student can describe in own words the concepts of radiative transfer, the greenhouse effect and the role of water and clouds on global climate
4. Student can describe in own words the fundamental drivers of tropical and extra-tropical circulation and its implications for weather
5. Student can describe in own words the importance of non-linear dynamics in the climate and how this can lead to abrupt transitions
6. Student has created a basic climate model of a 'Daisy' world and analyzed its dynamics
7. Student can critically analyze and interpret climate-related reporting in the media
8. Student can apply the new knowledge in class in presentations, discussions and debates.

Course Content

This course consists of several sessions consisting of (1) lectures by the professors with interactive discussion; (2) some practical assignments/coding, and (3) student presentations & discussions. You will team up in pairs of two (maybe three) students to develop a Daisy world climate model and do a series of experiments with it. Three afternoons will be devoted to this python coding where students will develop a basic climate model and perform some experiments. Also students will present in larger groups (~4/5) a recent scientific paper (or set of papers) which discusses the state-of-the-art of a particular lecture-topic. During these presentation sessions, all students are expected to participate in discussions.

Additional Information Teaching Methods

Class lectures (20hr), Computer practicals (12hr)

Method of Assessment

The course will be assessed through a written closed book examination based on the compulsory readings and the lectures. Also questions related to insights learned during the computer exercises can be asked.

There may be some quantitative question where a small calculation has to be made, so please bring a calculator. This closed book examination counts for 80% of the final grade. Furthermore, students will present a scientific paper in groups and discuss this in class. The presentation in class will together constitute 20% of the final grade.

The computer exercises are not graded but they are, however, a compulsory part of the course and need to be handed in via Canvas, for students to obtain a 'pass'.

Students will not receive a final grade if not all computer exercises have been handed in.

Literature

Apart from attending the sessions, you are expected to study the readings associated with the lectures ('compulsory reading') before attending them. This helps you both to actively engage in discussions during our meetings, and also makes it easier to prepare for the final exam at the end of the course.

The readings are provided as chapters and papers and aim to give you a broad understanding of the climate system as befits a master's program. However, you will be examined only on some elements of the course; in other words you do not have to memorize entire pieces of text. The structure of the lectures in class will help you identify the key questions on which you will be examined.

Recommended background knowledge

In the computer practicum Daisy World, you will work together in groups of 2 or 3 to build a simple simulation of Daisy World in python. If you do not have any experience in python, please complete the modules up to and including 'Functions' from the 'Learn the Basics' chapter on https://www.learnpython.org/

Please do so before the first practicum (normally in week 3 of the course)

Climate Modelling

Course Code	AM_450004
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. D.M.V.A.P. Roche
Examiner	prof. dr. D.M.V.A.P. Roche
Teaching Staff	dr. P. Bakker, prof. dr. D.M.V.A.P. Roche
Teaching method(s)	Seminar, Computer lab

Course Objective

The objective of this course is to provide an overview of numerical climate models and their applications, with a focus on Earth Science studies.

Course Content

Geological archives show convincingly that the climate system experiences variability on a wide range of time-scales. For Quaternary studies, climate variations at the following time-scales are most important: glacials-interglacials, millennia and centuries-decades. This course focuses at the mechanisms behind these variations, thereby using climate models as a tool, i.e. numerical computer models in which the dynamics of the climate system are calculated. The combination of these models and geological data will be treated extensively. The course consists of lectures giving an overview of climate models and their application (different types for different time-scales), computer practicals and discussion meetings, in which students discuss the recent literature in detail. In this way the course considers case studies for the different time-scales and deals with recent developments in climate modelling. The following two questions are central to the course:

1. What is the driving mechanism behind climate change at a particular time-scale?
2. How can we optimise the combination of climate models and geological data in order to increase our understanding of climate evolution?

Additional Information Teaching Methods

Lectures, discussion meetings and computer exercises.

Method of Assessment

Compulsory participation in discussion meetings, computer exercises, oral presentation and written exam.

Literature

- Text book: Goosse, H. (2015) Climate System Dynamics and Modelling. Cambridge University Press, 358 p., ISBN 978-1-107-44583-3

- Additional: Lecture notes and selected papers (made available through Canvas).

Additional Information

The course is open for participation to students from alternative M.Sc. programmes at VU University Amsterdam, or from other universities.
If you are a professional and wish to attend this course you can also participate on a contract basis. In both cases please do contact the course coordinator to find out if you fulfill the background knowledge requirements and for enrollment procedures.

Ecohydrology

Course Code	AM_450014
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. K. Naudts
Examiner	dr. K. Naudts
Teaching Staff	dr. K. Naudts
Teaching method(s)	Seminar, Computer lab, Lecture

Course Objective

Ecohydrology is a combination of ecology (study of how organisms interact with each other and with the natural environment) and hydrology (study of how water cycles in terrestrial environments). It focuses on the role of ecosystems in the water cycle of terrestrial landscapes. The objectives of the course is to provide understanding of the functioning of ecosystems in relation to water availability and the movement of water in terrestrial ecosystems under different climates. This ecohydrological knowledge forms the basis for supporting decisions on sustainable land use from a water resources point of view. It requires fundamental theoretical knowledge on plant physiology and on the exchange of water between the soil, vegetation and the atmosphere. As such, limitations to ecosystem functioning posed by water availability in relation to evaporation and transpiration by different plant communities is a central theme in this course.

Course Content

This course describes and discusses basic interactions between the vegetated land surface, the atmosphere and the hydrosphere. Basic questions dealt with include: what determines the broad vegetation patterns of the world, and how do these in turn determine the ecohydrological behaviour of different vegetation types? This requires understanding primary ecohydrological processes (rainfall interception, transpiration, soil moisture dynamics) and feedback mechanisms between the vegetation and the atmosphere as well as insight into the measurement, data analysis and modelling of these processes. Tropical and temperate deforestation impacts on catchment hydrological functioning and climate as well as desertification
processes are considered. Ecohydrological processes in boreal and tundra regions, as well as in montane cloud forests will be discussed in some detail. Emphasis throughout the course is on a combination of process understanding, interpretation of experimental results, and modelling. Finally, a workshop is included to become familiar with the basics of meteorological data processing, plant water flow and evaporation modelling.

Additional Information Teaching Methods

The tuition consists of nine classroom lectures, student presentation sessions, two workshops and a proposal writing assignment.

Method of Assessment

Written test on lecture notes and selected literature (65%), paper presentation assignment (15%), and a proposal writing assignment (20%).

Entry Requirements

The student should be familiar with the subjects of the BSc course Introduction to Hydrology (450024) as detailed in the Introduction to Hydrology (2012) .

Literature

Scientific papers and handouts are provided during the course via Canvas

Additional Information Target Audience

First-year MSc Hydrology students, students from alternative Earth Sciences, Earth and Economy or Natural Sciences MSc programmes

Additional Information

The course is open for participation to students from alternative M.Sc. programmes at the VU University Amsterdam, or from other universities.
If you are a professional and wish to attend this course you can also participate on a contract basis. In both cases please do contact the course coordinator to find out if you fulfill the background knowledge requirements and for enrollment procedures.

Recommended background knowledge

The student should have a good background knowledge of mathematics and physics at BSc level and basic computer skills

Environmental Policy for GEC&P

Course Code	AM_1241
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.C. Schoderer
Examiner	dr. M.C. Schoderer
Teaching Staff	dr. M.C. Schoderer
Teaching method(s)	Study Group, Lecture

Course Objective

After completing this course, the student is able to:

- Understand the concepts of governance and policy as they apply to challenges related to global environmental change;
- Distinguish between various instruments and approaches to environmental policy-making;
- Assess the effectiveness, legitimacy, and broader impacts of environmental policies;
- Perform an analysis of policy interventions and their logics;
- Present concrete science-based recommendations for decision-makers.

Course Content

Action to address environmental problems – i.e. environmental governance – nowadays does not only come from nation states but also from a plethora of other actors. In this course, you will learn what environmental policy and governance is and how some of the actors that provide for it operate. Specifically, you will learn to understand how environmental governance and policy is developed, implemented, assessed and contested at multiple levels, from domestic environmental policies to international treaties and transnational governance arrangements. You will familiarize yourself with the various modes of governance, from public to private, and the different governance instruments, from command and control to information-sharing and networking. In more detail, the course is thematically divided to address (a) the emergence and basic concepts of environmental policy and governance; (b) the role of interstate actors in environmental policy and governance (hierarchies); (c) the role of market-based approaches in environmental policy and governance (markets); (d) the role of networked governance with a focus on non-state actors (networks).

Additional Information Teaching Methods

Lectures

Method of Assessment

The course has the following required deliverables/assessments:

- Research papers (60%)
- Video presentation (40%)

Students must pass all elements (5.5 or higher).

Literature

Readings will be made available via Canvas.

Additional Information Target Audience

Students in the GEC&P specialization of MSc Earth Sciences.

Environmental Remote Sensing

Course Code	AM_450145
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S.S.N. Veraverbeke
Examiner	dr. S.S.N. Veraverbeke
Teaching Staff	dr. S.S.N. Veraverbeke, prof. dr. ir. S. Houweling
Teaching method(s)	Seminar, Computer lab, Written partial exam, Lecture

Course Objective

The course objectives are twofold:

1. Gaining an in-depth understanding of the theory of several remote sensing research applications
2. Executing an independent remote sensing project

Course Content

Lectures, including guest lectures, will cover specialized remote sensing research topics, for example on photogrammetry, surface water, fires, soil moisture, aerosols, greenhouse gases, urban environments, etc. The course also includes a project assignment, in which students will execute an independent remote sensing project.

Additional Information Teaching Methods

Lectures, including guest lectures, supplemented with reading materials.
Individual project with consultation sessions.

Method of Assessment

Written exam and individual project assignment, including poster presentation.

Literature

Selected research papers.

Additional Information Target Audience

First-year MSc Hydrology students and students from MSc Earth Sciences or other Natural Sciences MSc programs.

Recommended background knowledge

It is highly recommended that students have followed the course AB_1271 GIS en Aardobservatie in the VU BSc programs, or a similar course at another university.
The materials of this BSc course are assumed to be known, including practical experience with GIS and object-

oriented coding (preferably Python).

Field Research Project Hydrology

Course Code	AM_1314
Credits	12
Period	P5+6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. K. Naudts
Examiner	dr. K. Naudts
Teaching Staff	dr. H. de Moel, dr. K. Naudts
Teaching method(s)	Seminar, Computer lab, Fieldwork

Course Objective

This course main objective is to instill “hydrological system thinking” in the student's mind. This is done through a combination of practical application of the earth scientific and hydrologic theory given in the period before the field course to solve hydrological questions in the field. The objectives include planning and making decisions about research strategies (this includes defining testable hypotheses), learning to make relevant measurements of all the components of the hydrological cycle (surface water, ground water storage, soil moisture, vegetation, and the atmosphere) and use these measurements to make realistic qualitative and quantitative interpretations regarding the hydrological processes and conditions in the field and their relation to local and regional issues related to water resources. On top of that, the students learn how to organize and store their gathered data in a database. The course contributes to the final attainment levels Application of knowledge and insight; Critical judgement and Communication

Course Content

This course covers the practical side of hydrological research through application of geological and hydrological knowledge to solve the water balance of target areas and to study water quality issues. In this course, the students will carry out a research project that includes field work in Luxembourg. Focus will be on developing research skills, including making research questions and work plans, carrying out field measurements, processing data and critically discussing the whole process. The fieldwork course will consist of an introductory part at VU, then fieldwork in Luxemburg, followed by a three-week period after arrival back in Amsterdam to finalize the data processing and the report and to store and describe their data on a repository. During the first weeks of the course, lectures and practicals are given at the VU. During these lectures, you will get an introduction about the fieldwork area and you will be divided into small groups that will be assigned a study catchment. The other lectures and practicals focus on research skills that are needed for the fieldwork and for developing the work plan. In-between lectures, you will be working with your group on your work plan. After these first weeks, we will go to Luxembourg for almost 4

Additional Information Teaching Methods

The course is subdivided in three parts. Before the fieldwork preparatory lectures will be organised. In the fieldwork region, each group will be assigned a study catchment in which a hydrological observation network (surface water, ground water, meteorology, etc.) will be installed. Students are expected to work independently and make their own decisions regarding planning and research strategy. Data processing, analysis and modelling are an integral part of the field course to scale up the measurements and link their findings to water resources issues in the region. An individual report will be finalized in Amsterdam during the final three weeks of the course. Prior to the final version, the students are required to send in a draft report which will be reviewed by three peers (each student has to review three draft reports as well). Staff members will be present during the whole course period for supervision and for consultation by students.

Method of Assessment

Data management (10%)
Final presentation in the field (30%)
Publication of report on results (60%)

Entry Requirements

Admission to this field course is granted to students who have been admitted to the Hydrology MSc Programme. Furthermore, students must have completed the course Measuring Techniques in Hydrology and, before mid-April,

must have passed at least two of the courses Catchment Response Analysis, Groundwater Processes and Water Quality.

Literature

Handbook for Hydrological Field Measurements. Adapted from: Boris M. van Breukelen, Michel M.A. Groen, Koos Groen, Ko van Huissteden, Richard A.M. de Jeu, Vincent E.A. Post, Jaap Schellekens and Maarten J. Waterloo (2018). VU University Amsterdam.

Additional Information Target Audience

First year MSc Hydrology Programme students

Additional Information

The course coordinator will send you an e-mail asking for information about your participation in this course in January. Besides registering for this course via the VUnet portal please respond to the e-mail request of the coordinator. The course is partly subsidized by the faculty and the students are obliged to pay for the other part of the course (travel, residence costs, etc.).

Recommended background knowledge

The student should have a good general knowledge of the subjects discussed in the basic theoretical courses M.Sc. Hydrology master, i.e. Catchment Response Analysis, Ecohydrology, Groundwater Processes, Water Quality, Unsaturated Zone and Near Surface Hydrological Processes. Participants will need to work with GIS for analysing and displaying spatial spatial data and will need to be familiar with field methods.

Frontiers in Hydrology

Course Code	AM_1280
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M. Langer
Examiner	dr. M. Langer
Teaching Staff	dr. J.E. Vonk, dr. K. Naudts, dr. W.R. Berghuijs MSc, dr. ir. Y. van der Velde
Teaching method(s)	Seminar, Computer lab, Lecture

Course Objective

The Frontiers in Hydrology course is designed to explore the critical role that water plays in the Earth system, and to delve into the scientific and natural frontiers where new discoveries are being made. Through this course, we will examine a wide range of topics that cross the boundaries of climate regions, spatial scales, and Earth system components.

Specifically, we will provide an overview of the current state of scientific research in the field, including a review of literature, and a discussion of the publication process.

In lectures and practical exercises we will immerse ourselves in the latest hydrologic research being conducted at the VU and at the forefront of hydrologic science, in order to gain a deeper understanding of this exciting and rapidly evolving field.

In addition, we will learn how to plan and design our own small research project. By the end of this course, the participants should be equipped with the knowledge, skills, and resources needed to write short research proposals in the field of hydrology.

Course Content

Frontiers in Hydrology delves into the complex interactions between water and other components of the Earth system. In this course, we aim to understand how water connects different parts of the Earth, including the atmosphere, pedosphere, biosphere, and cryosphere.

Our exploration of hydrology is thereby not limited to physical processes; we also examine how water interacts with vegetation, and how it affects nutrient and carbon cycles. Our investigations cover a range of scales, from local to

regional, and we strive to push the boundaries of our understanding of hydrology by studying the interactions that occur across Earth system boundaries.

By examining these interactions, we aim to gain a deeper understanding of the role that water plays in our planet's complex and interconnected systems.

The course is composed of several elements, including:

- Literature review presentations by course participants. The presentations will give students the opportunity to share their insights and perspectives on current research in the field of hydrology.
- A series of lectures and practical exercises that cover the latest research and methods in the field. These lectures and exercises will be reinforced by weekly assignments, which will allow students to apply their knowledge and skills in a practical context.
- A final assignment, which will require students to develop and formulate their own small research project. This project will be presented to the class and will serve as the basis for a research proposal that students will complete in the form of a written paper.

Additional Information Teaching Methods

Lectures: 2 times 2 hours per week, Hands on computer/lab/field practice 2 times 2 hours per week. Presence is mandatory. Additional 12 hours per week working on assignments.

Method of Assessment

Proposal writing 40%, Presentations 30%, Assignments 30%

Entry Requirements

30ECTS in the Master Hydrology, Earth sciences, or environmental study program

Additional Information Target Audience

All interested students that followed at least 30ECTS of a relevant Master program

Global Biogeochemical Cycles

Course Code	AM_450332
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	W. Heffernan
Examiner	W. Heffernan
Teaching Staff	W. Heffernan, prof. dr. ir. S. Houweling, dr. J.E. Vonk
Teaching method(s)	Computer lab, Lecture

Course Objective

To better understand the importance of biogeochemical cycles in the Earth system, with specific attention to the role of biogeochemistry in climate change including for example ocean acidification, nitrogen and CO2 fertilisation, and impacts on society.

Course Content

Biogeochemical cycles are crucial for mankind and the Earth system. For example, they govern our food supply, regulate climate on long time scales based on the balance between weathering rates and volcanic eruptions, on short time scales via emissions of fossil fuels and climate-carbon feedbacks in oceans and permafrost regions, etc. The main subject of this course is the exchange of carbon, nitrogen, and phosphorus between the soil, water, atmosphere, and biota on global and local scales in different climatic zones (tropics, temperate, boreal and arctic zone) and environments. Each week consists of two lectures focused around a theme which will be followed by a practical where you will work with actual data to better understand links between biogeochemical cycles and the Earth system. The themes include the global terrestrial and oceanic carbon cycle and its relation to climate; the terrestrial nitrogen cycle and links to food production and pollution as well as climate; forests including deforestation and forest fires; carbon and nitrogen cycles in Arctic ecosystems and wetlands with a special focus on permafrost

dynamics; and oceanic cycles of nitrogen, carbon, and phosphorus. All these cycles are linked through the atmosphere which will be another theme during the course.

Additional Information Teaching Methods

14 Lectures, 7 practicals that include group work, discussion, and presentations

Method of Assessment

Evaluation of the practicals, student participation in group work, student engagement during class discussions, written exam, and possibly an essay

Literature

Lecture notes and scientific papers will be made available during the course.

Additional Information Target Audience

MSc students Earth Sciences, Hydrology, Environment and Resource management

Recommended background knowledge

We will use the Python scripting language during practicals and elemental knowledge of scripting is required (a manual will be provided), although some exercises can also be done using Excel

Groundwater Processes

Course Code	AM_1164
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ir. Y. van der Velde
Examiner	dr. ir. Y. van der Velde
Teaching Staff	dr. ir. Y. van der Velde
Teaching method(s)	Seminar

Course Objective

The objective of the course on Groundwater Processes is to gain knowledge and insight in the terminology and theory of groundwater hydraulics, including its mathematical notation and physical meaning.

Course Content

The movement of groundwater through the subsurface is a fundamental part of the hydrological cycle. In this course, you will get acquainted with the fundamental hydraulics of groundwater flow. You will get profound insight into fundamental hydrological concepts related to groundwater flow and its mathematical notation and solutions. The application of this knowledge will be illustrated using some basic groundwater flow modelling exercises.

Additional Information Teaching Methods

The course consists of a set of lectures supplemented with practicals.

Method of Assessment

Written examination

Additional Information Target Audience

Hydrology MSc students and other earth sciences related MSc programs

Integrated Modeling in Hydrology

Course Code	AM_1165
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. H. de Moel
Examiner	dr. H. de Moel
Teaching Staff	dr. S. Muis, dr. ir. D.M. Eilander, dr. H. de Moel
Teaching method(s)	Seminar, Computer lab

Course Objective

The objectives of this course are for the students to:

- get acquainted with the wide range of modeling tools used in integrated hydrological studies
- understand the type of questions that can be answered using such tools
- acquire hands-on modelling skills and gain experience with some commonly used analytical programs (Python, GIS, Excel)

Course Content

The course is set up with a limited amount of lectures and two large exercises. In these exercises the students will set up and apply their own models from start to finish. The first exercise concerns a spatial rainfall-runoff model using publicly available data sources. This model will be used to simulate river discharge under current and future climatic conditions, and results will be related to water resources availability and associated measures. The second exercise is a flood risk assessment where hazard, exposure and vulnerability will be
combined to estimate flood risks. Using the model, measures will be evaluated on their risk reducing effect.

Additional Information Teaching Methods

There will be a limited amount of lectures (in mornings) as the focus is on the development of the two exercises. Practical sessions are scheduled throughout the period where staff will be available to help with technical questions and to help start up the model development. The rest of the time students will work on their models, analyses and reporting themselves.

Method of Assessment

Assessment will be done in the form of a report (50%) and a poster presentation (50%) related to the assignments. Both grades need a minimum of 5.0 in order to pass the course.

Literature

Relevant course reading material (papers, reports) will be provided via Canvas.

Additional Information Target Audience

Hydrology MSc students and other earth sciences related MSc programs

Master Thesis Hydrology

Course Code	AM_1170
Credits	36
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. H. de Moel
Examiner	dr. H. de Moel
Teaching Staff	dr. S. Muis, dr. H. de Moel, dr. ir. Y. van der Velde
Teaching method(s)	Seminar

Course Objective

To conduct a research project in the scope of the MSc program Hydrology. The student will learn to develop his/her own research question, plan this research, collect data, perform analyses, draw proper findings and report on this.

Course Content

The student will set up, execute and report on a research project on a hydrological topic. This research and report serves as the final activity of the student within the MSc program where knowledge and skills obtained during the program are integrated and demonstrated by the student.

Additional Information Teaching Methods

This is course consist of self-study by the student, under guidance of a supervisor from the staff related to the MSc hydrology. There are monthly meetings planned throughout the 2nd year. These sessions are on progress (e.g. kick-off, mid-term, final presentations), as well as dedicated (soft) skills (e.g. writing, presenting, job interviews, etc.).

Method of Assessment

Written report/paper (60%), execution (30%) and oral presentation (10%).

Entry Requirements

A minimum of 36 ects is required before starting with this thesis project.

Additional Information Target Audience

MSc Hydrology students in the final stages of their study.

Measuring Techniques in Hydrology

Course Code	AM_1315
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M. Mazzoleni
Examiner	dr. M. Mazzoleni
Teaching Staff	dr. M. Mazzoleni, dr. P.J. Robinson, dr. S. Muis
Teaching method(s)	Seminar, Computer lab, Fieldwork, Practical

Course Objective

By the end of the course the student should be able:
1. to select the appropriate field measurement methods and techniques to measure hydrological processes and their values for society;
2. to apply and implement those methods and techniques, including equipment operation;
3. to analyze, evaluate and interpret the results, using computer models, statistical methods, and scientific literature;
4. to present key findings of the formulated hypothesis and data collected in the field.

Students from the MSc Hydrology should be able:
5. to carry out the final integrating fieldwork in Luxembourg and MSc thesis research projects independently.

Course Content

This course aims to familiarize students with measurement methods and techniques from different disciplines, commonly used in hydrology and related environmental sciences. The course is organized around measuring different hydrological processes, which are taught interlinked over a period of four weeks. The students are taught various measuring techniques used to quantify the hydrological processes. This includes methods to measure discharge, rainfall, evapotranspiration, soil properties, water quality, groundwater flow, and economic values. Students will learn how to select the appropriate method, conduct measurements, analyze and present the results. The measurements will be linked to hydrological theories and will be used and understand the main hydrological

processes in a region. During the first weeks of the course, practicals and lectures (e.g. on advanced topics such as Machine Learning, uncertainty, and Earth Observations) are given at the VU campus. After this, there is a one-week fieldwork in the area of the Dinkel River in Twente. During the fieldwork, the students will be taught how to apply measurement techniques in the field. Note that the fieldwork can take place during the holidays and weekend days and that during that week you will have to work in the evenings and cannot leave (unless there are special circumstances). Back at the VU campus, there is time to analyze the measurements taken in the field and to prepare the final presentation.

Additional Information Teaching Methods

Lectures, computer practicals, field practical, presentations

Method of Assessment

- Assignments related to the lectures and computer practicals
- Participation during field practical and presentations in the field
- Final presentation

Literature

Literature will be provided on Canvas.

Additional Information Target Audience

This course is compulsory for MSc Hydrology students. Students from other relevant MSc programs are allowed to join, but it is recommended to coordinate this with the course coordinator

Additional Information

The field course brings additional costs for the students, This is partly subsidized by the faculty, but the students are obliged to pay for part of the costs (travel, breakfast, lunch). Details on the will be shared during class.

Custom Course Registration

During the fieldwork you will work in groups, which will be assigned on Canvas.

Recommended background knowledge

It is recommended that students have a some general knowledge of hydrological processes. In addition to the field measurements, students will use GIS, Python and R for data analysis.

Plastic and Chemical Pollution

Course Code	AM_1259
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S.H. Brandsma
Examiner	dr. S.H. Brandsma
Teaching Staff	dr. S.H. Brandsma, dr. R. Oldenkamp, dr. L.A. Baumann, dr. S.L. Hughes, dr. I. van der Veen
Teaching method(s)	Computer lab, Seminar, Study Group, Symposium, College case, Practical, Lecture

Course Objective

Objectives

- To evaluate key environmental pollutants, their sources, properties, distribution and fate in the environment.
- To demonstrate quantitative information on the effects of pollutants on ecological and human health.
- To demonstrate how environmental risks of pollutants are assessed.
- To relate the concept of systems thinking, and its application to studying and understanding environmental pollution problems.
- To describe the key principles of environmental science that underpin both the assessment and governance

of plastic and chemical pollution.
- Formulate the environmental risk of a chemical pollutant in a report and an oral presentation.

Contents

This course provides an introduction to environmental chemistry and (eco)toxicology of plastic and chemical environmental pollution, one of the hallmarks of the hotly-debated ‘Anthropocene’, the human-dominated epoch. Is this environmental pollution just a global bundle of bad news, or can/does science and society adequately respond? If you are interested in learning about plastic pollution and other hot chemical pollution topics this course is for you.

Course Content

Topics include:

- Sources and emissions and spatial transport of plastic and chemical pollution
- Properties affecting transport and bioaccumulation of plastic and chemical pollutants
- Bioaccumulation and biotransformation in living organisms
- Ecotoxicology and human toxicology of selected plastic and chemical pollutants
- Risk assessment of plastic and chemical pollutants
- Systems thinking in environmental chemistry and one world chemistry
- Societal and policy responses to address pollution

Plastic and chemical pollution is dispersed throughout our environment, including food chains and human bodies, making it a problem of global proportions. But a solvable problem. It may be surprising but most of the >100,000 chemicals synthesized in the past 70 years have not undergone rigorous safety testing before entering the marketplace, and many of these are high-production volume chemicals, including polymers.

Approximately 80% of today’s chemicals sector produces plastic and polymers. Past lessons with pollutants like dichlorodiphenyltrichloroethane (DDT), polychlorinated biphenyls (PCBs), and chlorofluorocarbons (CFCs) showed us that chemicals with economic and societal benefits can have long-lasting serious negative consequences for public health, ecosystems, as well as the economy. After decades of research on an often chemical-by-chemical basis, science has been revealing how certain chemicals act as developmental neurotoxicants, endocrine disruptors, genotoxicants or immunotoxicants. And this knowledge is powerful. Such research is instrumental in evidence-based, risk-based policymaking that phased out many persistent organic pollutants to date. Cutting-edge analytical environmental chemistry is critical for first signaling and then quantifying ‘emerging’ pollutants in the environment, such as microplastics, endocrine disrupting chemicals (incl. plastic additive chemicals), perfluorinated chemicals, chlorinated paraffins, etc. Without analytical data, there’s no story to tell. Understanding the toxicity of the chemicals to humans and ecosystems gives meaning to the exposure concentrations measured in the field and laboratory. These data are essential input to environmental and human health risk assessments, policy-making and legislation, and design choices during manufacturing (e.g. green chemistry substitutions and functional alternatives). The societal and policy responses pave the way for elimination of pollution sources and the stimulation of innovations that act to prevent pollution, improving the quality of the living environment for all. This course may stimulate you to think about whether or not we currently are using our best science in creating the chemical-material world in which we are living.

Additional Information Teaching Methods

Teaching method

- Lectures
- Seminars
- Presentation
- Self-study
- Working on a written project
- Supervision/feedback

Method of Assessment

The final grade is calculated from the grade of the exam (65%), assignment report (25%), and oral presentation (10%). Minimum grades needed are: exam 5.0, report and presentation 5.5. In case resubmission of the report is needed a maximum grade of 6.0 will be assigned.

Entry Requirements

BSc degree in either hydrology, ecology, environmental sciences, chemistry, biomedical sciences, or public health.

If you have already followed the BSc Analytical and Environmental Chemistry (AEC) course (51128ANM6Y) then be aware that there can be overlap between some of the lectures given in this course.

Literature

Literature

“Environmental toxicology, an open online textbook”

Other: Additional material available on the Canvas site.

Additional Information Target Audience

Is part of/may be part of/is open to students of

- MSc Ecology (elective)
- MSc Hydrology (elective)
- MSc Earth Sciences – track Global Environmental Change and Policy (elective)
- Earth Sciences’ Earth & Climate stream (elective)
- MSc Biomedical Sciences (elective)
- MSc Chemistry (elective)

Additional Information

Attendance at lectures is highly recommended and is mandatory for the presentations.

The course is given at the VU. The first 4 weeks of the course will cover most of the new material, the second half of the course will be more focused on assignment, readings, a symposium for presentations of the assignment work, and finally the exam. Ecology students may complete work on assignment in the first four weeks, with only their presentation and exam in the second half of the course schedule.

Recommended background knowledge

BSc degree in either hydrology, ecology, environmental sciences, chemistry, biomedical sciences, or public health.

If you have already followed the BSc Analytical and Environmental Chemistry (AEC) course (51128ANM6Y) then be aware that there can be overlap between some of the lectures given in this course.

Reflection Seismic for Geologists

Course Code	AM_450170
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. K.F. Kuiper
Examiner	prof. dr. K.F. Kuiper
Teaching Staff	prof. dr. K.F. Kuiper, dr. A. Beniest, dr. T.S. Gunatilake
Teaching method(s)	Seminar, Computer lab, Lecture

Course Objective

Understanding the fundamentals of the application of reflection seismic to describe geology in the subsurface at depths of 200 m - 6 km. This includes:

- Seismic Sequence Stratigraphic interpretation
- Seismic Structural interpretation
- Seismic Acquisition and processing principles
- Calibration with petrophysical information/synthetic seismograms
- Time Depth Conversion methods
- Modern Techniques, Volume interpretation, Use of Attributes
- Examples/Special Case studies (4D time lapse, rock-property prediction)

During an extensive workstation Exercise (A/B), you will gain hands-on experience with subsurface interpretation. Here you will apply the theory to a real 3D seismic dataset, using an industry interpretation software package. The Exercise part of the course is done in teams of 2 or 3 persons. Teamwork is essential in this type of work and will also be gauged during the assessment.

Course Content

This course is part of the MSc Geology and Geochemistry stream "Subsurface Resourcing" which contains a series of elective courses that prepares for activities in the subsurface (geothermal energy, CO2 sequestration (CCS), hydrocarbon exploration and production). Many examples of successful techniques, originally developed for the oil and gas industry, are discussed. These techniques are nowadays often used in geothermal energy industry. Here you will learn how to use reflection seismic to interpret the subsurface and to make estimates of uncertainties that should be taken into account for a safe use of the subsurface.

Assuming a basic knowledge of the principles of reflection seismology, this course provides a modular program with principles and interpretation of reflection seismic data and geology, integrating data from wells and gaining hands-on experience in interpreting 3D seismic data. Special attention will be paid to pitfalls in data acquisition, processing and interpretation. During the course standard methodologies used in the industry will be applied.

The course consists of:

- Introduction to seismology. The introduction will cover the technical and methodological principles of reflection seismology. Also attention will be paid to the recognition and possible treatment of issues in seismic data quality and reliability.
- Structural interpretation. Students will learn how to interpret basic geological features, such as strata relationships, faults and folds as well as the reliability of seismic interpretation at various scales;
- Seismic Sequence Stratigraphy entails the extraction of stratigraphic, sedimentological and basin evolution information from seismic data. This information can be used in exploration and basin analysis to derive regional analysis of sedimentary basin-fills in order to construct models for gross lithology prediction (Students should review the principles of sequence stratigraphy, acquired during their BSc courses);
- Interpretation on a workstation. Students work on a case study using standard workstation methodologies and learn how to handle, visualize and interpret 2D and 3D seismic data using a standard industrial software package;
- Advanced seismic interpretation. This section will show examples of recently developed techniques in 3D seismic information analysis.

Additional Information Teaching Methods

The course uses two different methods:

- Oral lectures, where the lecturer presents various topics. Students must be aware that the content of this course is difficult to find in textbooks. Therefore, understanding the handouts is essential. Attendance to all oral lectures is compulsory.
- Workstation Exercises A and B. The practicals are done in teams of 2 or 3 students. Attendance of the practicals is compulsory. By the end of the course, students present the results of their team work in an oral presentation.

Method of Assessment

The final mark is made up by 50% from the result of the exam which will cover the topics presented during course. The oral presentation of the workstation exercise contributes another 50 % to the final score.

Literature

All materials will be digitally provided through Canvas.

Additional Information

Teaching staff: Guest lecturers.

Water Economics

Course Code	AM_1167
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. P.J. Robinson
Examiner	dr. P.J. Robinson
Teaching Staff	prof. dr. W.J.W. Botzen, prof. dr. P.J.H. van Beukering, dr. P.J. Robinson

Teaching method(s)	Seminar, Computer lab, Practical, Lecture

Course Objective

The objective of this course is to familiarise students with the economic analysis of water-related problems and solutions. After following this course, students should be able to judge how well certain policy instruments and institutional arrangements perform in terms of effectiveness, efficiency and the sustainability of water management. More specifically, after having participated in this module, students should be able to answer the following questions:

1. What is the fundamental nature of water related problems from an economic perspective, in relation to notions like externalities, public goods and free riding, and what does this imply for the choice of policy instruments?
2. How do we characterise and measure economic values of water?
3. What economic policy instruments are available for water management, and what are their (dis)advantages?
4. How do we guide the choice and design of water management strategies with the help of cost-benefit analyses, and what are the critical assumptions that underlie these methods?
5. What are the guiding principles of economic analyses of solutions for water related problems?

Course Content

Water is an essential natural resource for society and the economy. Every day people and various sectors of society, such as agriculture and industry, use large amounts of water for drinking, irrigation, food processing, transport, cooling, electricity production and recreation. Water is also a complex asset: it is renewable, recyclable and exhaustible at the same time. Water quality and quantity are equally relevant: having too little water makes water highly valued, whereas having too much results in a negative value of water. The real societal costs and benefits of water use often remain hidden from the public at large. Moreover, water can pose economic risks which are reflected in damages and economic production losses caused by floods and droughts.

The course Water Economics aims to introduce non-economics students to the analysis of the drivers behind water-related problems and their impacts on the environment and society. The course also deals with solutions regarding how to move towards a more sustainable and economically efficient use and management of water resources. Topics include water demand and supply, water markets and water pricing, valuation of water resources as well as costs and benefits of alternative water policies, e.g. when dealing with water pollution, river and groundwater ecosystem restoration, water quantity and water-related hazards such as floods. Moreover, integrated river-basin management will be addressed. The course offers both lectures and assignments in which students apply economic methods such as theoretical economic analyses, cost-benefit analysis (CBA) and meta-analysis, and within which they get familiar with statistical analysis using R.

Additional Information Teaching Methods

Lectures, Practicum (Computer labs).

Method of Assessment

Written exam and (group) assignments.

Literature

Readings will be announced on Canvas.

Additional Information Target Audience

Hydrologist, Water Science.

Water Politics and Governance

Course Code	AM_1192
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J. Dell'Angelo
Examiner	dr. J. Dell'Angelo

Teaching Staff	dr. J. Dell'Angelo
Teaching method(s)	Lecture

Course Objective

Water is central for ecosystems, human wellbeing and development. The relationship between society and water resources is critical and complex. Water issues have been historically treated as a technical problem, something to be resolved through hydraulic and engineering solutions. However, the increasing and exacerbating competition over water resources due to expanding global population, climate change and conflicting production activities raises new political and democratic challenges. Understanding water governance implies developing multidisciplinary knowledge about the different problems, approaches, perspectives and controversies related to how societies cooperate and compete over water resources. This course aims at building students' knowledge, critical thinking and analytical skills on the governance of freshwater resources with particular emphasis on the institutional, political and organizational dimensions.

Course objectives:

The main objective of this course is for you to develop an expert understanding of the multidimensional challenges of governance of water resources. Collectively as a class, and individually, you will develop expertise in the water governance world by engaging with the core scientific literature, discussing and tackling real world problems and independent research activities. By the end of the course, you will have gained an understanding of the complexity of water governance: the various models of governance, its structures and institutions at multiple scales, the different values and uses of water, and the main frames and debates from multiple perspectives.

By the end of this course you will be able to:

1. Describe various principles and theories for water governance and □management
2. Apply these to analyze different water policy and governance interventions, by analyzing them from diverse and diverging perspectives
3. Formulate your own analysis and suggestions regarding water-governance interventions on multiple scales: local, national and international
4. Synthesize and communicate your ideas and expertise orally and in written form

Course Content

Tentative program which can be revised, main themes are:

- Water politics and critical perspectives
- Neoliberalism and privatization
- Commons Theory and Community Irrigation Schemes
- Water Law and Water Rights
- NEXUS, Virtual Water, Water & Energy, Water and Mining
- International cooperation and water conflicts
- Global virtual water trade, water grabbing, water war and peace

Additional Information Teaching Methods

This course will be intensive, during three weeks you will be "immersed" in water governance studies. We will spend an important amount of the course in the classroom but you will be primarily required to study the readings to prepare for lectures and discussions. We will have three weekly meetings and each day we will mix: frontal lectures, discussions and group activities.

The structure of each session, generally, consists of lectures at the start, where we will first highlight concepts and key questions related to the topics we are working on. We will then engage in-group discussions associated with the key issues addressed in each session. The group activities will include simulations and team assignments and presentations. Readings will be assigned for each session and it will be essential to be well prepared in order to successfully participate and address key questions that we will tackle in the discussions. Finally, writing exercises, group activities, and debates will be an integral part of the course too.

Participation and preparation: Due to the complexity of the issues addressed and the concentrated length of the course I highly recommend attending every session. Of course, if you will have serious family or health reasons you can count on our help for making up for your absence and catching up, but otherwise it will really not be in your interest to miss classes. You are kindly invited inform the teaching assistant in case you cannot attend one session. The amount of the material that we will go through and the very short time to prepare between the end of classes and the final exam will not allow you to be successful in this course if you procrastinate.

We will have group discussions on key questions related to the readings lead by myself and other teaching assistants. In the discussions and lectures we will call on students directly to address specific question during the

group discussion. Referring to the “Socratic method” you might be challenged to defend your informed opinion and perspective on specific subjects. (This is a method that might seem a little intimidating, but is particularly useful to develop the capacity to debate publicly and defend your opinions – in line with the learning goals of the course).

Preparation advice: As MSc students you are very qualified, motivated and brilliant therefore some of this advice might be unnecessary. However, I can’t stress enough the importance to hit the ground and start running from day one, actively participate and not fall behind...keep in mind it’s only 3 weeks of classwork. A critical aspect is that you make an effort to study the assigned readings before class. If you don’t you will see pages adding up quickly and you not be as able to engage productively in discussions and have less information and tools to understand the lectures.

Method of Assessment

- Class participation, public debated, discussion and presentations.
- Water documentary review: This is a team project that includes a. search and selection of water documentaries; b. a written review of one selected documentary; c. a presentation to the class.
- In class final exam: there will be a closed-book 2.15h digital exam prevalently with open ended questions.

Literature

Selected articles and book chapters available in Canvas, organized around the above topics. There is no textbook for this class. For the most part, we will use scientific peer reviewed articles and, in certain cases, policy reports or other kind of documents. As we will be using scientific articles I will provide instructions on what I expect you should be focusing.

There will also be an additional number of suggested readings for the people interested in learning more about specific topics. Also, please note that the reading list available here in some rare cases might change during the course but you will all be clearly notified. You will find the final readings list every week and all the required readings in weekly folders on Canvas.

Additional Information Target Audience

Master level students with an interest in understanding the role of governance in the emergence of water issues, and their potential resolution

Additional Information

The course is coordinated and taught by Dr. Jampel Dell'Angelo (Associate Professor of Water Governance and Politics) with invited guest speakers. Language of teaching is English.

Custom Course Registration

Through the regular VU channels

Recommended background knowledge

no specific background required, just an eager motivation to engage, debate, learn and READ (a lot)

Water Risks

Course Code	AM_1210
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. ir. A.F. van Loon
Examiner	prof. dr. ir. A.F. van Loon
Teaching Staff	prof. dr. ir. A.F. van Loon, dr. J.A. de Bruijn
Teaching method(s)	Seminar, Lecture

Course Objective

The objective of this course is to provide students with a thorough understanding of concepts, datasets, methods, and applications in flood and drought risk assessments and management.

By the end of the course, students should be able to:

1. Review the scientific literature in order to discuss consecutive / co-occurring / compound floods and droughts, including hazard, exposure, vulnerability, and adaptation.
2. Discuss flood and drought risk reduction measures and strategies in case studies and propose improvements to current disaster risk assessment and management.
3. Describe the main datasets and methods (e.g. conceptual, statistical, computer simulation models) used to represent flood and drought hazard, exposure, and vulnerability, and explain their pros and cons.
4. Apply the main datasets and methods used to assess flood and drought hazard, exposure, and vulnerability.
5. Integrate data and information on hazard, exposure, and vulnerability to analyse flood and drought risk.

Course Content

Risk is defined as the probability of a natural hazard multiplied by its consequences, and can be seen as the product of the hazard, exposure, and vulnerability. In this course, students will learn the main concepts of water risk, how these water risks can be managed, and how these fit into international agreements on disaster risk reduction and recovery. They will also learn about different indicators that can be used to express flood and drought risk, and how flood and drought risk have changed over the past half century and are projected to change in the 21st century. Students will learn about the main datasets and methods that can be used to assess hazard, exposure, and vulnerability, and apply several of these to assess flood and drought risk at different spatial scales. Finally, students will learn about different measures that can be taken to reduce flood and drought risk, and the pros and cons of these measures.

Additional Information Teaching Methods

In weeks 1-6, the course will be taught as a mixture of (guest) lectures, practicals, discussions, and serious games. Sessions are interactive, so active participation of all students is required. We aim to develop an open and respectful atmosphere, in which diverse views can be expressed. A sit-in exam is taken in week 8.

Several sessions are mandatory (e.g., guest lectures) and for the other sessions attendance is strongly recommended. The sessions require preparation and/or work after the session of around three hours (per session). This consists of reading scientific papers, looking up information and working on practicals.

In order to participate in the course, students need to bring their own laptop computers and will need to be able to install and run the following programs: Python, SFINCS, Microsoft Office, QGIS. For Python and SFINCS a guide will be distributed before the course and assistance will be provided during the first lecture if needed.

Method of Assessment

The course will be assessed through 2 components: the results report of computer practicals (50%) and a sit-in exam (50%). The final grade is a weighted mean of the grades for the 2 components, whereby students must achieve a grade of at least 5.5 for each component. Resit opportunities will be provided in the form of: a re-take of the sit-in exam and/or improving the results report of the computer practicals.

Entry Requirements

Knowledge of GIS, basic programming and hydrological processes.

Literature

A reading list will be provided on CANVAS.

Additional Information Target Audience

MSc Hydrology, MSc Earth Sciences (Global Environmental Change & Policy, Earth & Climate), MSc Climate Econometrics, and others as long as they have knowledge of GIS, basic programming and hydrological processes.

Recommended background knowledge

In preparation for the course, students can find some guidance on learning Python and several mini-tutorials in last year's syllabus: https://vu-ivm.github.io/Water-Risks/mini_tutorials/learning-python.html. Note that the practical exercises may be updated before the start of the course.

To brush up on hydrological process knowledge, students can have a look at a hydrology textbook, such as "Hydrology: an introduction" by Brutsaert or "Principles of hydrology" by Ward & Robinson (both are available in the VU library).